

Homework 3

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```
In [1]: using Pkg
Pkg.add("Plots")
using Plots
Pkg.add("JuMP")
using JuMP
Pkg.add("HiGHS")
using HiGHS
Pkg.add("DataFrames")
using DataFrames
Pkg.add("CSV")
using CSV
Pkg.add("NamedArrays")
using NamedArrays
Pkg.add("LinearAlgebra")
using LinearAlgebra
Pkg.add("XLSX")
using XLSX
Pkg.add("PrettyTables")
using PrettyTables
```

[illegible]

Question 1-1

Decision Variables:

C_t , Money Barry has at year t

b_i , bonds of type i bought

s_i , bonds of type i sold

Objective:

$$\max C_3$$

Subject To:

$$C_t \geq 0 \quad \forall t \in T$$

$$1000 \geq b_i \geq 0 \quad \forall i \in I \text{ and } \forall t \in T$$

$$1000 \geq s_i \geq 0 \quad \forall i \in I \text{ and } \forall t \in T$$

$$C_0 = 100000 - \sum_{i=1}^4 (\text{Ask Price}_i \cdot b_i) + \sum_{i=1}^4 (\text{Bid Price}_i \cdot s_i)$$

$$C_1 = 1.03 \cdot C_0 + \sum_{i=1}^4 (\text{Payout}_{i,1} \cdot (b_i - s_i)) - 10000$$

$$C_2 = 1.03 \cdot C_1 + \sum_{i=1}^4 (\text{Payout}_{i,2} \cdot (b_i - s_i)) - 11000$$

$$C_3 = 1.03 \cdot C_2 + \sum_{i=1}^4 (\text{Payout}_{i,3} \cdot (b_i - s_i))$$

Question 1-2

In [2]: **using** JuMP, HiGHS

```
# Define sets
T = 0:3                                # Years
I = 1:4                                # Bond types

ask_price = [980, 970, 960, 940]      # Ask prices of bonds
bid_price = [990, 985, 972, 954]      # Bid prices of bonds
payout = [                             # Payouts per year for each bond
    [100, 110, 1100],
    [80, 90, 1120],
    [70, 80, 1090],
```

```

    [60, 50, 1110]
]
expenses = [0, 10000, 11000, 0]    # Expenses at each year
total_initial_cash = 100000        # Initial cash
interest_rate = 1.03               # 3% interest per year

model = Model(HiGHS.Optimizer)

@variable(model, C[T] >= 0)         # Cash at year t
@variable(model, 0 <= b[I] <= 1000) # Bonds bought
@variable(model, 0 <= s[I] <= 1000) # Bonds sold

@constraint(model, C[0] == total_initial_cash - sum(ask_price[i] * b[i] for
@constraint(model, C[1] == interest_rate * C[0] + sum(payout[i][1] * (b[i] -
@constraint(model, C[2] == interest_rate * C[1] + sum(payout[i][2] * (b[i] -
@constraint(model, C[3] == interest_rate * C[2] + sum(payout[i][3] * (b[i] -

@objective(model, Max, C[3])

# Solve the model
set_silent(model)
optimize!(model)

# Display results
println("Optimal final cash: ", objective_value(model))
println("Bonds bought:")
for i in I
    println("Bond $i: ", value(b[i]))
end

println("Bonds sold:")
for i in I
    println("Bond $i: ", value(s[i]))
end

```

Optimal final cash: 222261.00425531916

Bonds bought:

Bond 1: 1000.0

Bond 2: 1000.0

Bond 3: 0.0

Bond 4: 80.85106382978724

Bonds sold:

Bond 1: 0.0

Bond 2: 0.0

Bond 3: 1000.0

Bond 4: 1000.0

Question 1-3

Decision Variables:

C_t , Money Barry has at year t

b_i , bonds of type i bought

$s_i^{\leq 500}$, first 500 bonds of type i sold

$s_i^{> 500}$, last 500 bonds of type i sold

Objective:

$$\max C_3$$

Subject To:

$$C_t \geq 0 \quad \forall t \in T$$

$$1000 \geq b_i \geq 0 \quad \forall i \in I \text{ and } \forall t \in T$$

$$500 \geq s_i^{\leq 500} \geq 0 \quad \forall i \in I \text{ and } \forall t \in T$$

$$500 \geq s_i^{> 500} \geq 0 \quad \forall i \in I \text{ and } \forall t \in T$$

$$s_i = s_i^{\leq 500} + s_i^{> 500}$$

$$C_0 = 100000 - \sum_{i=1}^4 (\text{Ask Price}_i \cdot b_i) + \sum_{i=1}^4 (\text{Bid Price}_i^{\leq 500} \cdot s_i^{\leq 500}) + \sum_{i=1}^4 (\text{Bid Price}_i^{> 500} \cdot s_i^{> 500})$$

$$C_1 = 1.03 \cdot C_0 + \sum_{i=1}^4 (\text{Payout}_{i,1} \cdot (b_i - s_i)) - 10000$$

$$C_2 = 1.03 \cdot C_1 + \sum_{i=1}^4 (\text{Payout}_{i,2} \cdot (b_i - s_i)) - 11000$$

$$C_3 = 1.03 \cdot C_2 + \sum_{i=1}^4 (\text{Payout}_{i,3} \cdot (b_i - s_i))$$

Question 2-1

Decision Variables:

S_t : number of snitches to produce during period t

I_t : The increase in production rates between periods t - 1 and t

D_t : The decrease in production rates between periods t - 1 and t

O_t : Total inventory level at end of period t

$N_t^{\leq 8000}$: Number of snitches (up to 8000) at the end of period t

$N_t^{> 8000}$: Number of snitches (over 8000) at the end of period t

Cost:

$$C = \sum_{t=July}^{December} (1.50 \cdot I_t + 1.00 \cdot D_t + 0.10 \cdot S_t + 0.20 \cdot N_t^{\leq 8000} + 0.50 \cdot N_t^{>8000})$$

Objective:

$$\min C$$

Subject To:

$$0 \leq N_t^{\leq 8000} \leq 8000$$

$$0 \leq N_t^{>8000}$$

$$O_t = O_{t-1} + S_t - \text{demand}_t$$

$$O_t = N_t^{\leq 8000} + N_t^{>8000}$$

$$S_t = S_{t-1} + I_t - D_t$$

$$S_t, I_t, D_t, O_t, N_t^{\leq 8000}, N_t^{>8000} \geq 0$$

Question 2-2

```
In [3]: T = [:june, :july, :august, :september, :october, :november, :december]
demand = Dict{:july => 4000, :august => 8000, :september => 20000, :october

model = Model(HiGHS.Optimizer)

@variable(model, S[T] >= 0)
@variable(model, I[T] >= 0)
@variable(model, D[T] >= 0)
@variable(model, O[T] >= 0)
@variable(model, NL[T] >= 0)
@variable(model, NG[T] >= 0)

@constraint(model, S[:june] == 4000)
@constraint(model, O[:june] == 2000)

@constraint(model, [t in T], NL[t] <= 8000)
@constraint(model, [t in T[2:end]], O[t] == O[T[findfirst(==(t), T)-1]] + S[t]
@constraint(model, [t in T], O[t] == NL[t] + NG[t])
@constraint(model, [t in T[2:end]], S[t] == S[T[findfirst(==(t), T)-1]] + I[t]

@objective(model, Min, sum(1.50 * I[t] + 1.00 * D[t] + 0.10 * S[t] + 0.20 *

set_silent(model)

optimize!(model)
```

```

println("Termination status: ", termination_status(model))
println("Cost: \$", objective_value(model))
println("Optimal production plan:")
for t in T[2:end]
    println("$t: Produce ", value(S[t]), " snitches, Inventory: ", value(O[t])
end
for t in T[2:end]
    println("Month: ", t)
    println("  S (Production): ", value(S[t]))
    println("  I (Increase): ", value(I[t]))
    println("  D (Decrease): ", value(D[t]))
    println("  O (Inventory): ", value(O[t]))
    println("  NL ( $\leq 8000$ ): ", value(NL[t]))
    println("  NG ( $> 8000$ ): ", value(NG[t]))
    println("-----")
end

```

Termination status: OPTIMAL

Cost: \$25266.666666666664

Optimal production plan:

july: Produce 10000.0 snitches, Inventory: 8000.0

august: Produce 10666.666666666666 snitches, Inventory: 10666.666666666666

september: Produce 10666.666666666668 snitches, Inventory: 1333.333333333332
1

october: Produce 10666.666666666668 snitches, Inventory: -0.0

november: Produce 8000.0 snitches, Inventory: 2000.0

december: Produce 8000.0 snitches, Inventory: 8000.0

Month: july

S (Production): 10000.0

I (Increase): 6000.0

D (Decrease): 0.0

O (Inventory): 8000.0

NL (≤ 8000): 8000.0

NG (> 8000): 0.0

Month: august

S (Production): 10666.666666666666

I (Increase): 666.6666666666661

D (Decrease): 0.0

O (Inventory): 10666.666666666666

NL (≤ 8000): 8000.0

NG (> 8000): 2666.666666666666

Month: september

S (Production): 10666.666666666668

I (Increase): 0.0

D (Decrease): 0.0

O (Inventory): 1333.3333333333321

NL (≤ 8000): 1333.3333333333321

NG (> 8000): 0.0

Month: october

S (Production): 10666.666666666668

I (Increase): 0.0
 D (Decrease): 0.0
 O (Inventory): -0.0
 NL (≤ 8000): 0.0
 NG (> 8000): 0.0

Month: november

S (Production): 8000.0
 I (Increase): 0.0
 D (Decrease): 2666.666666666668
 O (Inventory): 2000.0
 NL (≤ 8000): 2000.0
 NG (> 8000): 0.0

Month: december

S (Production): 8000.0
 I (Increase): 0.0
 D (Decrease): 0.0
 O (Inventory): 8000.0
 NL (≤ 8000): 8000.0
 NG (> 8000): 0.0

Question 2-3

Decision Variables:

S_t : number of snitches to produce during period t

I_t : The increase in production rates between periods t - 1 and t

D_t : The decrease in production rates between periods t - 1 and t

O_t : Total inventory level at end of period t

$N_t^{\leq 8000}$: Number of snitches (up to 8000) at the end of period t

$N_t^{> 8000}$: Number of snitches (over 8000) at the end of period t

B_t : Total amount backlogged at end of period t t

Cost:

$$C = \sum_{t=July}^{December} (1.50 \cdot I_t + 1.00 \cdot D_t + 0.10 \cdot S_t + 0.20 \cdot N_t^{\leq 8000} + 0.50 \cdot N_t^{> 8000} + 0.25 \cdot .$$

Objective:

$$\min C$$

Subject To:

$$0 \leq N_t^{\leq 8000} \leq 8000$$

$$0 \leq N_t^{>8000}$$

$$O_t = O_{t-1} + S_t - \text{demand}_t + B_{t-1} - B_t$$

$$O_t = N_t^{\leq 8000} + N_t^{>8000}$$

$$S_t = S_{t-1} + I_t - D_t$$

$$\sum_{t=\text{July}}^{\text{December}} S_t = \sum_{t=\text{July}}^{\text{December}} \text{demand}_t$$

$$B_{\text{December}} = 0$$

$$S_t, I_t, D_t, O_t, N_t^{\leq 8000}, N_t^{>8000}, B_t \geq 0$$

```
In [4]: T = [:june, :july, :august, :september, :october, :november, :december]
demand = Dict{:july => 4000, :august => 8000, :september => 20000, :october

model = Model(HiGHS.Optimizer)

@variable(model, S[T] >= 0)
@variable(model, I[T] >= 0)
@variable(model, D[T] >= 0)
@variable(model, O[T] >= 0)
@variable(model, NL[T] >= 0)
@variable(model, NG[T] >= 0)
@variable(model, B[T] >= 0)

@constraint(model, S[:june] == 4000)
@constraint(model, O[:june] == 2000)
@constraint(model, B[:december] == 0)

@constraint(model, [t in T], NL[t] <= 8000)
@constraint(model, [t in T[2:end]], O[t] == O[T[findfirst(==(t), T)-1]] + S[t])
@constraint(model, [t in T], O[t] == NL[t] + NG[t])
@constraint(model, [t in T[2:end]], S[t] == S[T[findfirst(==(t), T)-1]] + I[t])
@constraint(model, sum(S[t] for t in T[2:end]) == sum(demand[t] for t in key

@objective(model, Min, sum(1.50 * I[t] + 1.00 * D[t] + 0.10 * S[t] + 0.20 *

set_silent(model)

optimize!(model)

println("Termination status: ", termination_status(model))
println("Cost: \$", objective_value(model))
println("Optimal production plan:")
```

```

for t in T[2:end]
    println("$t: Produce ", value(S[t]), " snitches, Inventory: ", value(O[t])
end
for t in T[2:end]
    println("Month: ", t)
    println("  S (Production): ", value(S[t]))
    println("  I (Increase): ", value(I[t]))
    println("  D (Decrease): ", value(D[t]))
    println("  O (Inventory): ", value(O[t]))
    println("  NL ( $\leq 8000$ ): ", value(NL[t]))
    println("  NG ( $> 8000$ ): ", value(NG[t]))
    println("-----")
end

```

Termination status: OPTIMAL

Cost: \$19400.000000000004

Optimal production plan:

july: Produce 8666.666666666666 snitches, Inventory: 6666.666666666666

august: Produce 8666.666666666666 snitches, Inventory: 7333.333333333332

september: Produce 8666.666666666666 snitches, Inventory: -0.0

october: Produce 8666.666666666666 snitches, Inventory: -0.0

november: Produce 8666.666666666666 snitches, Inventory: -0.0

december: Produce 8666.666666666666 snitches, Inventory: 1999.9999999999964

Month: july

S (Production): 8666.666666666666

I (Increase): 4666.666666666666

D (Decrease): 0.0

O (Inventory): 6666.666666666666

NL (≤ 8000): 6666.666666666666

NG (> 8000): 0.0

Month: august

S (Production): 8666.666666666666

I (Increase): 0.0

D (Decrease): 0.0

O (Inventory): 7333.333333333332

NL (≤ 8000): 7333.333333333332

NG (> 8000): 0.0

Month: september

S (Production): 8666.666666666666

I (Increase): 0.0

D (Decrease): 0.0

O (Inventory): -0.0

NL (≤ 8000): 0.0

NG (> 8000): 0.0

Month: october

S (Production): 8666.666666666666

I (Increase): 0.0

D (Decrease): 0.0

O (Inventory): -0.0

NL (≤ 8000): 0.0

NG (> 8000): 0.0

Month: november

S (Production): 8666.666666666666
I (Increase): 0.0
D (Decrease): 0.0
O (Inventory): -0.0
NL (≤ 8000): 0.0
NG (> 8000): 0.0

Month: december

S (Production): 8666.666666666666
I (Increase): 0.0
D (Decrease): 0.0
O (Inventory): 1999.9999999999964
NL (≤ 8000): 1999.9999999999964
NG (> 8000): 0.0

Question 3-1

Decision Variables:

$$x_t \quad \forall t \in \{1, 2, 3, 4, 5, 6\}$$

Constraints:

$$3x_1 + 3x_2 + 4x_3 + 6x_6 = 15$$

$$2x_1 - 6x_2 - x_3 - 5x_4 + 2x_5 - x_6 = 10$$

$$x_1 - x_2 + x_3 - 3x_5 + x_6 = 2$$

$$2x_1 + x_3 - x_4 + 6x_5 = 0$$

Objective:

$$\min \sum_{i=1}^6 |x_t|$$

Question 3-2

```
In [5]: model = Model(HiGHS.Optimizer)

@variable(model, x[1:6])
@variable(model, y[1:6] >= 0)

@constraint(model, y[1:6] >= x[1:6])
@constraint(model, y[1:6] >= -x[1:6])

@constraint(model, 3*x[1] + 3*x[2] + 4*x[3] + 6*x[6] == 15)
@constraint(model, 2*x[1] - 6*x[2] - x[3] - 5*x[4] + 2*x[5] - x[6] == 10)
@constraint(model, x[1] - x[2] + x[3] - 3*x[5] + x[6] == 2)
```

```

@constraint(model, 2*x[1] + x[3] - x[4] + 6*x[5] == 0)

@objective(model, Min, sum(y[t] for t in 1:6))

set_silent(model)
optimize!(model)

println("Termination status: ", termination_status(model))
println("Optimal solution found:")
for t in 1:6
    println("x_$t = ", value(x[t]))
    # println("y_$t = ", value(y[t]))
end

```

Termination status: OPTIMAL

Optimal solution found:

x_1 = -1.4109589041095891

x_2 = -0.0

x_3 = -0.0

x_4 = -3.232876712328767

x_5 = -0.06849315068493148

x_6 = 3.2054794520547945

Optimal solution found:

x_1 = -1.4109589041095891

x_2 = -0.0

x_3 = -0.0

x_4 = -3.232876712328767

x_5 = -0.06849315068493148

x_6 = 3.2054794520547945

Question 3-3

Decision Variables:

$$x_t \quad \forall t \in \{1, 2, 3, 4, 5, 6\}$$

Constraints:

$$3x_1 + 3x_2 + 4x_3 + 6x_6 = 15$$

$$2x_1 - 6x_2 - x_3 - 5x_4 + 2x_5 - x_6 = 10$$

$$x_1 - x_2 + x_3 - 3x_5 + x_6 = 2$$

$$2x_1 + x_3 - x_4 + 6x_5 = 0$$

Objective:

$$\min \max |x_t|$$

Question 3-4

```

In [6]: model = Model(HiGHS.Optimizer)

@variable(model, x[1:6])
@variable(model, y[1:6] >= 0)

@constraint(model, y[1:6] >= x[1:6])
@constraint(model, y[1:6] >= -x[1:6])

@constraint(model, 3*x[1] + 3*x[2] + 4*x[3] + 6*x[6] == 15)
@constraint(model, 2*x[1] - 6*x[2] - x[3] - 5*x[4] + 2*x[5] - x[6] == 10)
@constraint(model, x[1] - x[2] + x[3] - 3*x[5] + x[6] == 2)
@constraint(model, 2*x[1] + x[3] - x[4] + 6*x[5] == 0)

@objective(model, Min, y)

set_silent(model)
optimize!(model)

println("Termination status: ", termination_status(model))
println("Optimal solution found:")
for t in 1:6
    println("x_$t = ", value(x[t]))
    # println("y_$t = ", value(y[t]))
end

```

Termination status: DUAL_INFEASIBLE

Optimal solution found:

x_1 = 0.0
 x_2 = 0.42696629213483145
 x_3 = 0.4297752808988764
 x_4 = -0.5646067415730338
 x_5 = -0.1657303370786517
 x_6 = -0.5

Optimal solution found:

x_1 = 0.0
 x_2 = 0.42696629213483145
 x_3 = 0.4297752808988764
 x_4 = -0.5646067415730338
 x_5 = -0.1657303370786517
 x_6 = -0.5

Question 3-5

Decision Variables:

$$x_j \quad \forall j \in \{1, 2, 3, \dots, n\}$$

$$y_j \quad \forall j \in \{1, 2, 3, \dots, n\}$$

$$z_j \quad \forall j \in \{1, 2, 3, \dots, n\}$$

Subject To:

$$x_j = y_j - z_j \quad \text{with} \quad y_j, z_j \geq 0$$

$$\sum_{j \in N} a_{ij}(y_j - z_j) = b_i \quad \forall i \in M$$

$$y_j, z_j \geq 0 \quad \forall j \in N$$

Objective:

$$\min \sum_{j \in N} (y_j + z_j)$$

Question 4-1

Decision Variables:

t_t Start time of task $t \in T$

d_t Duration of task $t \in T$

$P = \{(i, j) : i \in T \text{ directly precedes } j \in T\}$

Constraints:

$$t_j \geq t_i + d_i \quad \forall j \in P_i \quad \forall i \in T$$

Objective:

$$\min_t t_{18} + d_{18}$$

Question 4-2

```
In [7]: tasks = 1:18

dur = [2, 16, 9, 8, 10, 6, 2, 2, 9, 5, 3, 2, 1, 7, 4, 3, 9, 1]
duration = Dict{zip(tasks,dur)}

pre = ( [], [1], [2], [2], [3], [4,5], [4], [6], [4,6], [4], [6], [9], [7],
pred = Dict{zip(tasks,pred)}

model = Model(HiGHS.Optimizer)

@variable(model, tstart[tasks])

@constraint(model, [i in tasks, j in pred[i]], tstart[i] >= tstart[j] + duration[i])
@constraint(model, tstart[1] == 0)
@objective(model, Min, tstart[18] + duration[18])

set_silent(model)
optimize!(model)
```

```
println("Termination Status: ", termination_status(model))
println(value.(tstart))
println("minimum duration: ", objective_value(model))
```

Termination Status: OPTIMAL

1-dimensional DenseAxisArray{Float64,1,...} with index sets:

Dimension 1, 1:18

And data, a 18-element Vector{Float64}:

```
-0.0
 2.0
18.0
18.0
27.0
37.0
26.0
43.0
43.0
26.0
43.0
52.0
28.0
18.0
26.0
46.0
54.0
63.0
```

minimum duration: 64.0

Question 4-3

Decision Variables:

t_t Start time of task $t \in T$

d_t Duration of task $t \in T$

P $\{(i, j) : i \in T \text{ directly precedes } j \in T\}$

r_t Weeks task t is completed early $\in T$

m_t Max \# of weeks task t can be reduced

c_t Cost per week to reduce task t

$\chi = 64$ Baseline time to complete the project

Constraints:

$$t_j \geq t_i + d_i - r_i \quad \forall j \in P_i \quad \forall i \in T$$

$$0 \leq r_i \leq m_i \quad \forall j \in P_i \quad \forall i \in T$$

Objective:

$$\max 30(\chi - (x_{18} + d_{18})) - \sum_{i \in T} (r_i \cdot m_i)$$

Question 4-4

```
In [8]: tasks = 1:18

dur = [2, 16, 9, 8, 10, 6, 2, 2, 9, 5, 3, 2, 1, 7, 4, 3, 9, 1]
duration = Dict{zip(tasks,dur)}

pre = ( [], [1], [2], [2], [3], [4,5], [4], [6], [4,6], [4], [6], [9], [7],
pred = Dict{zip(tasks,pre)}

max_reduction = [0, 3, 1, 2, 2, 1, 1, 0, 2, 1, 1, 0, 0, 2, 2, 1, 3, 0]
cost_per_week = [0, 30, 26, 12, 17, 15, 8, 0, 42, 21, 18, 0, 0, 22, 12, 6, 1

m_reduction = Dict{zip(tasks, max_reduction)}
c_cost = Dict{zip(tasks, cost_per_week)}

model = Model(HiGHS.Optimizer)

@variable(model, tstart[tasks] >= 0 )
@variable(model, r[tasks] >= 0)

@constraint(model, [t in tasks], r[t] <= m_reduction[t])
@constraint(model, [i in tasks, j in pred[i]], tstart[i] >= tstart[j] + dura
@constraint(model, tstart[1] == 0)
@objective(model, Max, 30*(64 - (tstart[18] + duration[18])) - sum(c_cost[i]

set_silent(model)
optimize!(model)
println(value.(tstart))
println("Maximun Profit: ", objective_value(model))
```

1-dimensional DenseAxisArray{Float64,1,...} with index sets:

Dimension 1, 1:18

And data, a 18-element Vector{Float64}:

0.0
2.0
18.0
18.0
26.0
34.0
26.0
39.0
39.0
26.0
39.0
48.0
28.0
18.0
26.0
42.0
50.0
56.0

Maximun Profit: 87.0

Dimension 1, 1:18

And data, a 18-element Vector{Float64}:

0.0
2.0
18.0
18.0
26.0
34.0
26.0
39.0
39.0
26.0
39.0
48.0
28.0
18.0
26.0
42.0
50.0
56.0

Maximun Profit: 87.0

Question 5-1

Decision Variables:

R : Set of restaurants

d_{ij} : Distance from restaurant i to restaurant j

P : Path taken (start, end)

x_{ij} 1 if the path from i to j is taken otherwise 0

Constraints:

$$\sum_j x_{ss,j} = 1$$

$$\sum_i x_{i,a} = 1$$

$$\sum_i x_{ik} = \sum_j x_{kj} \quad \forall k \neq \text{Short Stack, Alchemy}$$

Objective:

$$\min \sum_{(i,j) \in P} d_{ij} \cdot x_{ij}$$

Question 5-2

```
In [9]: restaurants_end = [ :gd, :of, :rs, :iag, :wt, :a]
restaurants_start = [:ss, :gd, :of, :rs, :iag, :wt]

# restaurants_y -> restaurants_x == OKAY
# restaurants_x -> restaurants_y == BAD
dist = NamedArray(
    [4      6      5      100000      100000      100000;
      100000      1      100000      7          100000      100000;
      100000      100000      2      5.5      4          100000;
      100000      100000      100000      100000      5          100000;
      100000      100000      100000      100000      1          6;
      100000      100000      100000      100000      100000      8],
    (restaurants_start, restaurants_end),
    ("restaurant", "restaurant")
)

model = Model(HiGHS.Optimizer)

@variable(model, x[restaurants_start, restaurants_end], Bin) # is the drive

@objective(model, Min, sum(dist[i, j] * x[i, j] for i in restaurants_start for j in restaurants_end))

@constraint(model, sum(x[:ss, j] for j in restaurants_end) == 1)
@constraint(model, sum(x[i, :a] for i in restaurants_start) == 1)
for k in setdiff(restaurants_start, [:ss, :a])
    @constraint(model, sum(x[i, k] for i in restaurants_start) == sum(x[k, j] for j in restaurants_end))
end

set_silent(model)
optimize!(model)
```

```

if termination_status(model) == MOI.OPTIMAL
    println("Optimal Solution Found!")
    println("Shortest Path length: ", objective_value(model))
    for i in restaurants_start
        for j in restaurants_end
            if value(x[i, j]) > 0.5
                println("Path from $i to $j")
            end
        end
    end
else
    println("No optimal solution found.")
end

```

```

Optimal Solution Found!
Shortest Path length: 16.5
Path from ss to gd
Path from gd to of
Path from of to iag
Path from iag to a
Shortest Path length: 16.5
Path from ss to gd
Path from gd to of
Path from of to iag
Path from iag to a

```